

STATISTICAL TESTS ON THE CONSISTENCY OF RESULTS OF DIFFERENT POLLS UNDER A TWO-VOTE SYSTEM

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Summary: In a unique block vote system where each voter is allowed to vote for two candidates in his constituency, tests are proposed for various sets of hypotheses on the consistency of results of polls conducted by different institutions. The meanings, implications and relationships of these hypotheses are discussed. The results of polls conducted by three academic institutions on the elections by geographical constituencies to the Legislative Council of Hong Kong in 1991 are analysed. The critical values of the test statistics can be closely approximated using chi-squared distributions. It is found that the tests give very similar results with the conclusion that the three institutions gave very different results statistically.

1. INTRODUCTION

In 1991, Hong Kong held its first direct elections by geographical constituencies to the Legislative Council. Many polls were conducted on these historic elections. Most of the polls were territory-wide, and were conducted by professional survey research companies or academic institutions. This article investigates whether or not the institutions obtained consistent results.

Three tertiary institutions, namely the Hong Kong Institute of Asia-Pacific Studies of the Chinese University of Hong Kong (AP), the Political Science Department of Lingnam College (PS), and the Social

Sciences Research Centre of the University of Hong Kong (*RC*), provided the poll data. The private companies did not release their data. There were nine double-seat geographical constituencies in the 1991 direct elections (for more detail, see Fung 1993). The polls involved a random sampling of voters who were contacted by telephone (over 95% of the households in Hong Kong have telephones). Respondents were asked to select up to two candidates in their respective constituencies. This block vote electoral system is quite unique in electoral studies (Lakeman, 1970), and little discussion of the system is found in the statistical literature. As each voter could have up to two votes, the vote size was always larger than the voter size. A typical example of the election survey results in the New Territories-North (NT-N) constituency is given in table 1. Table 2 gives the summary information of the polls conducted in mid-August, 1991. The number of candidates per constituency ranged from four to eight. Of the three institutions, *RC* had the smallest sample size. This may have affected the statistical significance of the results.

TABLE 1. Poll results for the New Territories North (NT-N) constituency

	Candidates																	
	A			B			C			D			E			F		
A	4	4	2															
B	2	4	1	2	12	3												
C	3	9	2 ^a	3	9	3	1	3	1									
D	4	3	2	3	8	1	1	2	4	2	1	4						
E	2	1	0	1	10	5	3	4	1	1	1	1	0	1	1			
F	0	0	0	0	1	0	0	1	0	0	0	0	0	1	0	0	0	0

^a - 3, 9 and 2 persons voted for both candidates A and C (double-votes) in the *AP*, *PS* and *RC* polls respectively. The diagonal terms in the table are single votes.

TABLE 2. Summary information of the polls conducted in mid-August 1991

	NT-E ^a		NT-S		NT-W		NT-N		KL-E		KL-C		KL-W		HK-E		HK-W		Date conducted
k ^b	7		4		5		6		8		7		6		6		6		
	n ^c	m ^d	n	m	n	m	n	m	n	m	n	m	n	m	n	m	n	m	
AP	70	114	83	139	81	118	32	55	77	126	87	150	81	144	111	191	71	127	12–14 Aug.
PS	136	227	124	194	109	162	75	129	130	205	162	261	87	137	194	321	88	137	13–19 Aug.
RC	36	62	– ^e	–	83	131	31	51	64	91	63	96	50	81	97	146	41	63	14–15 Aug.

a - NT = New Territories, KL = Kowloon, HK = Hong Kong Island, E = East, S = South, W = West, N = North, and C = Central

b - k = number of candidates in the constituency

c - n = number of voters in the constituency selected by the poll

d - m = number of votes

e - the RC poll results for NT-S are missing

The marketing research companies and academic institutions like to claim that their surveys are objective, unbiased and representative. As the surveys were conducted at a similar period of time, it is of interest to see whether the results (table 1) obtained by different institutions are consistent with each other. This is investigated in section 2. As many cells in table 1 contain very small or zero frequencies, it is also suggested that the total votes obtained by candidates should be of concern. Section 3 studies the consistency of results under this formulation. It is found that the polls gave inconsistent (i.e. statistically significantly different) results, indicating a problem of validity of the poll findings. As the votes so obtained are not statistically independent, the critical values of the test statistics are evaluated under a Monte Carlo simulation. They are found to be well approximated using chi-squared distributions. The implication of the hypothesis of independence under this new formulation is also studied in section 3. Section 4 compares the results under the two formulations. We also examine the results under the more stable dropping and combining procedures (votes of unpopular candidates dropped or combined to remove irregularities); similar findings are obtained (section 3). A few concluding remarks are given in section 5.

2. RESULTS BY CANDIDATE PAIRS

It is of interest to see whether the three institutions gave consistent poll results for every constituency. This is done in the following way: table 1 can be rearranged to a standard 3×21 contingency table as in table 3. The usual tests for the null hypothesis of independent row and column factors of a contingency table can be used. In other words, we are interested in whether the polls gave consistent results of vote shares for candidate pairs. The null and alternative hypotheses are

H_0 : The three institutions gave consistent results of vote shares of candidate pairs, and

H_1 : H_0 is not true.

The popular test for independence in a contingency table is the Pearson's chi-squared test which is asymptotically χ^2 distributed. A common guideline for using the asymptotic result is that the expected frequencies should be at least five (Fienberg, 1979, suggested that smaller expected frequencies will suffice under some conditions), which definitely does not hold for table 3. The small sample distribution of the test statistic is discussed by Agresti (1990). The evaluation of the exact distribution involves heavy computational work, but the Monte Carlo simulation can give a very good approximation to the exact distribution. Another popular test for independence is Fisher's exact test (see Conover, 1980, for details). The Monte Carlo simulation is also used for approximating the exact significance level.

TABLE 3. Votes for candidate pairs for polls in the NT-N constituency (Re-arrangement of table 1 results)

	Candidate pairs					
	A, A	A, B	A, C	...	E, F	F, F
<i>AP</i>	4	2	3	...	0	0
<i>PS</i>	4	4	9	...	1	0
<i>RC</i>	2	1	2	...	1	0

Both Pearson's chi-squared and Fisher's exact tests are employed for testing independence for the NT-N constituency (table 3). A Monte Carlo simulation of 4 000 repetitions is run for each test using the StatXact software. The simulated p-values are respectively 0.666 and 0.568, with a width of about 0.04 for the 99% confidence intervals, for Pearson's and Fisher's tests. Both p-values are large, and the null hypothesis H_0 that the polls gave consistent results of vote shares for candidate pairs is accepted. The hypothesis of independence of poll results in other constituencies are

analysed in similar ways. The estimated p-values (from Monte Carlo simulations) are shown in the second and third columns of table 4. The maximum width of the 99% confidence intervals of the p-values is 0.04. Both Pearson's and Fisher's tests identify significant results for the same six (out of a total of nine) constituencies each at the 5% significance level. This conclusion remains unchanged even if the 99% confidence interval estimates of the p-values are employed. It seems that the results of the three institutions were generally inconsistent. As it is noticed that the two columns of p-values of table 4 agree well, so only the more popular Pearson's chi-squared test is used for the discussion below.

TABLE 4. The p-values under various tests for independence in contingency tables

	Candidate Pairs ^a		Individual Candidates ^b			
	F	P	P ₁	P ₂	P ₃	P ₄
NT-E	0.025	0.018*	0.131	0.135	0.054	0.054
NT-S	0.124	0.110	0.506	0.499	0.351	0.306
NT-W	0.001	0.002*	0.108	0.116	0.039	0.045*
NT-N	0.568	0.666	0.228	0.252	0.118	0.117
KL-E	0.000	0.001*	0.001	0.002	0.001	0.000*
KL-C	0.019	0.009*	0.075	0.037	0.015	0.011*
KL-W	0.010	0.009*	0.021	0.017	0.007	0.005*
HK-E	0.014	0.030*	0.207	0.124	0.047	0.048*
HK-W	0.092	0.242	0.101	0.161	0.074	0.067

a - p-values for hypotheses H_0 vs. H_1 using Fisher's (F) exact and Pearson's (P) chi-square tests

b - p-values for hypotheses H_0 vs. H_1 . P_1 and P_3 are obtained using simulations, P_2 and P_4 are obtained from χ^2 distributions with d_1 and d_2 df respectively under the dropping procedure. Both P_1 and P_2 are derived under the independent observations assumption

3. RESULTS BY INDIVIDUAL CANDIDATES

It may be argued that studying votes for *candidate pairs* is somewhat inappropriate because the election outcomes depend on the (total) votes obtained by *individual candidates*. The votes for the candidates in the NT-N constituency are given in table 5 which is also a contingency table. Another argument for considering votes for individual candidates is that the frequency counts in table 5 are much larger than those in table 3 which is a bit sparse. The results in table 5 are more stable.

Table 5 also contains a few cells having small frequencies (see results for candidate F). Similar tables for other constituencies exhibit the same phenomenon. Thus we may have to use Monte Carlo simulations, as in section 2, for obtaining the p-values of the Pearson chi-squared statistics. The hypothesis on the consistency of vote shares of candidates obtained by different institutions would be of concern here. The null and alternative hypotheses are now given as

H'_0 : The three institutions gave consistent results of vote shares of individual candidates, and

H'_1 : H'_0 is not true.

The p-values for the hypotheses, obtained using simulations with 4 000 repetitions each, are given in column 4 of table 4.

TABLE 5. Votes for candidates for polls in the NT-N constituency

	Candidates					
	A	B	C	D	E	F
AP	15	11	11	11	7	0
PS	21	44	28	15	18	3
RC	7	13	11	12	8	0

One crucial assumption for getting these p-values is that the observations in the contingency table are statistically independent. This, however, is not true because under the two-vote electoral system, each voter could vote for two candidates, and so the vote would be counted twice in the table. Thus, the "true" p-values may have to be evaluated using simulation, which involves considerable computational effort because double simulations (simulation here and simulation for Pearson's test for the sparse contingency table) are required.

The double simulations can be avoided using the procedure described below. The usual requirement for employing the asymptotic χ^2 distribution for Pearson's test statistic is that the expected frequencies are at least (about) five. This is not true for those of the last column (candidate F) of table 5. In similar tables of other constituencies, some meet the requirement, while in others one or two columns fail to meet it. We suggest dropping columns (candidates) which do not satisfy the requirement. This procedure thus ignores the votes of unpopular candidates. As it is believed that unpopular candidates have only very small chances of winning the elections, the irregularity of these unstable results should not affect our decision about consistency. This gives a justification for the suggested dropping procedure. Moreover, the procedure can be interpreted as testing for the consistency of results for more popular candidates.

The asymptotic χ^2 distribution with $d_1 = (r-1)(c-1)$ degrees of freedom (df), where r and c are the numbers of rows and columns under the dropping procedure, is used for determining the p-values. These p-values are given in column 5 of table 4. If a 5% significance level is taken, the results agree well with those obtained under Monte Carlo simulations (column 4), except in the KL-C constituency of which the p-values, 0.075 and 0.037, are both close to 0.05. It suggests that the dropping procedure only has little effect on the p-values.

Since the observations in table 5 are not independent as described, the p-values obtained above (columns 4 and 5 of table 4) are inappropriate. The distribution of Pearson's test statistic under the dropping procedure is

determined using simulation as follows:

The (population) distribution of votes for candidate pairs in constituency NT-N is obtained by combining the corresponding votes of the three institutions in table 3. A (simulated) sample of 32, 75 and 31 observations respectively for the institutions AP, PS and RC is randomly selected from the distribution. A table like table 5 is then constructed for the votes obtained by candidates. Hence, Pearson's chi-squared test statistic under the dropping procedure is evaluated. The simulation is repeated 4 000 times to determine the null distribution of the test statistic. The simulated ('exact') p-value of Pearson's statistic, 10.197, in the NT-N constituency is obtained as 0.118, which is much smaller than 0.252 (see table 4), determined under the independent observation assumption (using the χ^2 distribution with $d_1 = (r - 1)(c - 1)$ df as described before).

It would be valuable if a simple (approximate) distribution for Pearson's test statistic could be obtained. If the tail probability of the statistic 10.197 is evaluated under a χ^2 distribution with $d_2 = (r - 1)(c - 2)$ df, we obtain a probability 0.117 which is very close to the simulated p-value 0.118. Moreover, the simulated 0.1, 0.05 and 0.01 critical values respectively have tail probabilities 0.101, 0.057 and 0.013 under this χ^2 distribution. The p-values of the test statistics for the other eight constituencies are also determined by Monte Carlo simulations and by χ^2 distributions with d_2 degrees of freedom. They are given in columns 6 and 7 of table 4 respectively. They agree very well with each other. Thus, it seems that the χ^2 distribution with d_2 df gives a very good approximation to the simulated results, though the reason for the good approximation is still to be explored.

Simulations for cases in comparing any two institutions (i.e. $r = 2$ only), and for artificially constructed data sets have also been conducted. Based on all these (more than 40) simulations, we find that the simulated distributions can be well approximated by χ^2 distributions with $d_2 = (r - 1)(c - 2)$ df, especially at the upper tails of the distributions. The simulated upper 10% points are largely within 0.02 from the corresponding

tail probabilities of the χ^2 distributions with d_2 df.

Although exact methods are available for analysing sparse contingency tables, they often require heavy computational work. The approximation using the χ^2 distributions with d_2 df is generally appropriate for analysing data sets of the present block vote system in which most respondents make use of both their votes (see table 2). However, the χ^2 distribution with d_1 df is valid if all potential voters only use one of their two votes, but this extreme case almost never happens in practice.

The test, under the dropping procedure, applies to testing for independence of vote shares of *more popular* candidates in each constituency under different institutions. The original nominal test (without dropping columns) instead applies to *all* candidates in each constituency. Its null distribution can be evaluated but involves very heavy computational work of double simulations. Since the conservative p-values, obtained under the independent observation assumption, for the two tests are so close (columns 4 and 5 of table 4), we believe that the actual p-values for the nominal test can be well approximated by those of the test under the dropping procedure (column 7 of table 4). In other words, column 7 of table 4 can be interpreted as the p-values for the test on independence of vote shares of all or more popular candidates under different institutions.

Besides the dropping procedure, a combining procedure has also been examined. This procedure combines (instead of drops) columns until the expected frequencies of the contingency table are at least five. It removes the irregularity of the unstable results by combining votes of unpopular candidates. The resulting p-values are also very close to those obtained under the dropping procedure, so they are not reported here.

4. COMPARISONS AND IMPLICATIONS OF RESULTS

There are two sets of hypotheses H_0 vs. H_1 , and H'_0 vs. H'_1 considered in sections 2 and 3 respectively. Obviously, H_0 is a stronger statement than H'_0 . The following shows that $H_0 \Rightarrow H'_0$, but $H'_0 \not\Rightarrow H_0$.

Consider H_0 for table 3 which is 3×21 . Let p_{ij} be the probability that a randomly selected voter votes for candidate pair (column) j in survey (row) i , $i = 1, \dots, 3$, $j = 1, \dots, 21$, $p_{i\cdot} = \sum_j p_{ij}$, and $p_{\cdot j} = \sum_i p_{ij}$. Also define q_{ik} , $q_{i\cdot}$ and $q_{\cdot k}$, $i = 1, \dots, 3$, $k = 1, \dots, 6$ in similar ways for table 5, in which the hypothesis H'_0 is of interest. Table 5 is obtained from table 3 which itself is a rearrangement of table 1 results. Candidate A has votes in columns 1, 2, ..., 6 of table 3. These columns are denoted by the set $S_1 = (1, 2, \dots, 6)$. Similarly, with reference to tables 1 and 3, $S_2 = (2, 7, 8, 9, 10, 11)$, $S_3, \dots, S_6$ are respectively the sets of columns containing the votes for candidates B, C, ..., F. If we sum the probabilities for candidate A under survey i in table 3, it is $t_{i1} = \sum_{j \in S_1} p_{ij}$. The sum for other candidates B, ..., F are similarly obtained as $t_{ik} = \sum_{j \in S_k} p_{ij}$, $k = 2, \dots, 6$. It is noticed that $\sum_i \sum_j p_{ij} = \sum_i \sum_k q_{ik} = 1$, but $\sum_i \sum_k t_{ik} = W \neq 1$ because of the double counting. However, $q_{ik} = t_{ik} / W$.

The two null hypotheses can be reformulated as $H_0 : p_{ij} = p_{i\cdot} \times p_{\cdot j}$, for all i and j , and $H'_0 : q_{ik} = q_{i\cdot} \times q_{\cdot k}$, for all i and k . In the following, we show that if H_0 is true, H'_0 would be true too. Under H_0 , the three individual terms in H'_0 are given as,

$$(1) \quad q_{ik} = t_{ik} / W = \frac{1}{W} \sum_{j \in S_k} p_{ij} = \frac{1}{W} \sum_{j \in S_k} p_{i\cdot} \times p_{\cdot j} \\ = \frac{p_{i\cdot}}{W} \sum_{j \in S_k} p_{\cdot j},$$

$$(2) \quad q_{i\cdot} = \sum_k q_{ik} = \frac{p_{i\cdot}}{W} \sum_k \sum_{j \in S_k} p_{\cdot j} \\ = p_{i\cdot},$$

because $t_{\cdot k} = \sum_i p_{\cdot j}$ and $W = \sum_k t_{\cdot k}$, and

$$\begin{aligned}
 (3) \quad q_{\cdot k} &= \sum_i q_{ik} = \frac{1}{W} \sum_i p_{i\cdot} \sum_{j \in S_k} p_{\cdot j} \\
 &= \frac{1}{W} \sum_{j \in S_k} p_{\cdot j}
 \end{aligned}$$

because $\sum_i p_{i\cdot} = 1$. It is obvious that (1) = (2) $\times$ (3). That is, H_0 implies H'_0 . However, it is easy to find examples that H'_0 does not imply H_0 . Mathematically, $H_0 \Rightarrow H'_0$, or equivalently $\bar{H}_0' \Rightarrow \bar{H}_0$, i.e. $H'_1 \Rightarrow H_1$.

The p-values for the two sets of hypotheses, H_0 vs. H_1 , and H'_0 vs. H'_1 are given in columns 3 and 7 of table 4. If the significance level is set at 5%, H_0 is accepted in three constituencies (NT-S, NT-N and HK-W), of which H'_0 is also accepted. Conversely the data support the alternatives H'_1 , are accepted too. Thus, the findings are compatible with the theoretical results that $H_0 \Rightarrow H'_0$, and $H'_1 \Rightarrow H_1$ given above. Comparing the two sets of p-values, it is found that six constituencies have p-values smaller than 0.05 for H_0 vs. H_1 , giving the same results, except in the NT-E constituency, as those for H'_0 vs. H'_1 . The result for H'_0 vs. H'_1 in NT-E is only marginally insignificant. The findings under the two sets of hypotheses are similar.

The null hypotheses H_0 and H'_0 are rejected at six and five constituencies respectively. It seems that the three institutions gave rather inconsistent election results. A further step in the investigation involves combining the independent p-values. The null hypothesis now is that the polls gave consistent results in all constituencies. A common approach is Fisher's method (Snedecor and Cochran, 1956, § 9.4). The statistic, $X = -2 \sum_{i=1}^9 \log(p_i) \sim \chi^2_{18}$ under the null hypothesis, where p_i is the individual p-value for constituency i , $i = 1, \dots, 9$. The p-values for the X statistics are obtained as 6×10^{-8} and 2×10^{-7} for H_0 and H'_0 (columns 3 and 7 of table 4) respectively. They are remarkably close and are significantly small. Thus, they provide very strong evidence that the polls gave significantly different results.

5. CONCLUDING REMARKS

Various polls were conducted on the 1991 direct elections in Hong Kong. It is expected that they will continue to be used in future elections. As in Western countries, most of the polls were sponsored by the mass media and conducted by survey research companies and academic institutions. The results of these polls were widely covered in the press, but there was little concern for the survey methodologies. This article investigates the consistency of the results of three polls conducted during a similar period of time.

Two sets of hypotheses about consistency are formulated. For the hypothesis about the consistency of votes obtained by individual candidates, three procedures – the nominal, dropping and combining procedures – are tried. Although the institutions often claimed that surveys were reliable and representative, all these formulations have very similar findings with the conclusion that the polls gave significantly different results. Further analyses on pairwise comparisons indicate that *AP* and *PS* gave reasonably consistent results, but the *RC* results were different.

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